# WEEK 2 PROGRESS UPDATE

*Deepening the Study of Speech Models, Multilingual Understanding, and Indirect Intent*

Research Progress Report

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| **Reporting Period** | Week 2 | **Status** | Ongoing — Exploratory Phase |

## 1. Problem Understanding — Spoken Language Understanding (SLU)

My project falls under Spoken Language Understanding (SLU). SLU aims to understand the meaning and intention of a speaker from spoken input.

The two major components are:

- Intent Detection: Identifying what the speaker wants or intends.
- Slot Filling: Extracting important information/entities that help understand the intent.

Example:

“Book a flight from Hyderabad to Delhi tomorrow.”

Intent: Book Flight Slots: Hyderabad, Delhi, Tomorrow

## 2. Transformer-Based Models

Transformers became important in NLP after the introduction of the Transformer architecture in 2017. The major concept is self-attention, which allows the model to understand relationships between different words/tokens in a sequence.

The attention mechanism uses:

- Query (Q): What information is being searched for.
- Key (K): Information used to determine relevance.
- Value (V): Actual information passed forward.

Attention scores are calculated mathematically, and Softmax converts the scores into normalized attention weights. This allows the model to use surrounding words and context when understanding meaning.

## 3. Cascading Error

A common approach is:

**Audio**
↓
**Speech-to-Text**
↓
**Intent Detection**

The problem is that an error in speech recognition can propagate to the next stage. For example, an incorrect transcription leads directly to an incorrect intent. This is known as a cascading error. Therefore, relying only on the converted text can potentially discard useful information contained in the original audio.

## 4. Speech Models Explored

### 4.1 Whisper

Whisper is a Transformer-based speech model developed by OpenAI.

Main task: Audio → Text

It supports multilingual speech recognition and can also perform speech translation.

Relevance to my project: Whisper can be used as the speech-to-text component.

Possible pipeline:

**Audio**
↓
**Whisper**
↓
**Text**
↓
**Intent Detection**

An important consideration is whether to translate the speech into English. Translation may sometimes alter subtle meaning or intention, so retaining the original-language transcription and using a multilingual language model is another possible approach.

### 4.2 wav2vec 2.0

wav2vec 2.0 is a self-supervised speech representation learning model. Its main purpose is not direct intent detection or transcription. It learns useful representations from raw/unlabelled audio and these representations can then be used for downstream speech tasks.

Basic idea:

**Unlabelled Audio**
↓
**Self-supervised Learning**
↓
**Speech Representation**
↓
**Fine-tuning**
↓
**Downstream Task**

Relevance: It can provide information from the audio itself, rather than throwing away the acoustic information after transcription.

### 4.3 XLS-R

XLS-R (Cross-Lingual Speech Representation) is a multilingual speech representation model from Meta AI based on the wav2vec 2.0 family. It was developed to learn speech representations across many languages.

Main purpose: Multilingual Audio → Speech Representation

Relevance: Multilinguality is one of the major problems in my project, so XLS-R is an important candidate for the audio branch. It does not directly detect intent — it provides a speech representation that can be passed to a downstream intent model.

### 4.4 HuBERT

HuBERT = Hidden-Unit BERT.

It is another self-supervised speech representation model developed by Meta AI. It learns useful speech representations from unlabelled speech using masked prediction of hidden/pseudo speech units.

Similarity with wav2vec 2.0: Both are Transformer-based, self-supervised, speech representation models suitable for downstream fine-tuning.

Difference: They use different self-supervised learning objectives. HuBERT is therefore another possible candidate for the audio representation branch.

## 5. Multilingual Text Models

### 5.1 XLM-R

XLM-R = XLM-RoBERTa. It is a multilingual Transformer-based text model developed by Meta AI. Unlike Whisper, XLS-R and HuBERT, XLM-R works on text, not raw audio.

Possible pipeline:

**Audio**
↓
**Whisper**
↓
**Original-language Text**
↓
**XLM-R**
↓
**Intent**

XLM-R is pretrained, but to specifically perform intent detection, it can be fine-tuned using an intent-labelled dataset.

Relevance: XLM-R is particularly useful for the multilingual text → intent part of the project. It can potentially allow the system to process the original language instead of always translating everything into English.

### 5.2 LaBSE

LaBSE = Language-agnostic BERT Sentence Embedding. It was developed by Google. Its main purpose is to generate multilingual sentence embeddings.

For example, sentences with the same meaning in different languages can be represented in a shared semantic space.

Relevance: It can help with multilingual semantic representation, cross-lingual similarity, and multilingual intent classification when combined with a classifier.

Difference from XLM-R: XLM-R focuses on general multilingual contextual text understanding, while LaBSE is particularly focused on multilingual sentence-level semantic embeddings.

## 6. Role of Context

Context is information surrounding the current utterance, such as previous dialogue turns. Context becomes particularly important for indirect intent.

Example — without context:

“It's very hot here.” — the intended action is unclear.

Example — with context:

Person: “The fan isn't working.” Person: “It's very hot here.”

The intended meaning could be: Turn on the fan.

Therefore: Context → additional information → better understanding of indirect intent. Context can include:

- Previous utterances

- Dialogue history
- Dialogue state
- Previous questions/answers

## 7. Direct vs Indirect Intent

Direct intent — “Turn on the fan.” The intended action is explicitly stated.

Indirect intent — “It's very hot here.” The speaker doesn't explicitly say “Turn on the fan.” The model must infer the intended action. This makes indirect intent considerably more difficult.

Important research observation: Existing intent datasets contain many intent-labelled examples, but they are not necessarily designed specifically for indirect intent. Therefore, additional data/annotation may be required for explicitly studying indirect intent.

## 8. Datasets Explored

### 8.1 MInDS-14

A multilingual spoken intent dataset focused on banking. Contains spoken data, 14 language varieties, intent labels, and banking-related intents.

Relevance: Very relevant to multilingual speech → intent.

Limitation: It is domain-specific and does not completely solve multilingual + audio + text + context + indirect intent.

### 8.2 MASSIVE

A large multilingual SLU dataset containing approximately 1M+ utterances, 51/52 languages depending on version, 60 intents, and 55 slot types.

Relevance: Very useful for multilingual text → intent + slot filling, particularly for experimenting with models such as XLM-R.

Limitation: It does not provide the complete raw-audio + indirect-intent + context resource needed for my exact project.

### 8.3 Skit-S2I

An Indian-accented speech-to-intent dataset in the banking domain.

Relevance: Useful for studying speech-to-intent, Indian accents, speech representations, and problems with ASR-based approaches.

Limitation: It does not provide broad multilingual coverage required for my project.

### 8.4 Banking77

A well-known banking intent dataset containing 77 intent categories.

Relevance: Useful for intent classification, training/fine-tuning intent models, and baseline experiments.

Limitations: Primarily English, text-based, not specifically designed for indirect intent, and not a raw multilingual audio dataset.

### 8.5 Dataset Observation

There does not appear to be one single dataset that completely provides: Multilingual + Speech + Text + Acoustic Information + Context + Indirect Intent. Therefore, existing datasets can be treated as building blocks:

- MASSIVE → Multilingual text + intents + slots
- MInDS-14 → Multilingual spoken intent
- Skit-S2I → Indian-accented speech-to-intent
- Additional data/annotation → Indirect intent + context

## 9. Possible Approaches for the Project

### Approach 1 — Audio → Text → Intent

**Audio**
↓
**Whisper**
↓
**Text**
↓
**Intent Model**
↓
**Intent**

Simple and practical, but susceptible to cascading ASR errors.

### Approach 2 — Audio → Speech Representation → Intent

**Audio**
↓
**XLS-R / wav2vec 2.0 / HuBERT**
↓
**Speech Representation**
↓
**Intent Model**
↓
**Intent**

This preserves information from the audio that may be lost during transcription.

### Approach 3 — Audio + Text → Intent

**Whisper**
↓
**Text**

**XLS-R**
↓
**Audio Representation**

**Intent**

This is particularly interesting because the system does not completely discard the acoustic information.

### Approach 4 — Audio + Text + Context → Intent

**Whisper**
↓
**Text**

**XLS-R**
↓
**Audio Representation**

**Fusion Layer + Context**

**Intent Detection**

This is currently a promising architecture to investigate, especially for indirect intent. However, it should be experimentally validated rather than assumed to be the best solution.

## 10. Main Problems Identified

### A. Multilinguality

Challenges include multiple Indian languages, different accents, code-mixing, low-resource languages, and different ways of expressing the same intent. For example, the same intention can be expressed differently in Telugu, Hindi, Tamil, or English.

### B. Indirect Intent

The literal meaning of a sentence may not equal the intended action. For example, “It's very hot here” may possibly mean “Turn on the fan.” This requires deeper semantic/pragmatic understanding.

### C. Cascading Errors

**Audio**
↓
**ASR Error**
↓
**Incorrect Text**
↓
**Incorrect Intent**

This motivates investigating audio information directly instead of depending entirely on ASR.

### D. Dataset Limitation

Existing datasets usually solve some of the requirements, but not all simultaneously. Therefore, creating or extending a dataset specifically containing multilingual + indirect intent + context may become part of the research.

## 11. Current Research Direction

A promising direction identified so far is:

Combine multilingual speech information, textual information, and conversational context to improve intent detection, especially for indirect intents.

A possible architecture is:

**Whisper**
↓
**XLS-R**

**Multilingual Text**
↓
**Audio Features**

**Fusion + Context**

**Intent Classifier**

This is one candidate architecture, not yet the finalized solution.

## 12. Current Research Gap

Existing research has addressed multilingual intent detection, spoken intent detection, speech representations, and context-aware intent detection individually or in combinations. However, a system/dataset that simultaneously handles multilingual speech, acoustic information, textual information, conversational context, and indirect intent is much more limited.

Therefore, the project can investigate whether combining these sources of information improves intent detection, particularly for multilingual and indirect intents.

## Conclusion

Week 2 built on the initial exploration by studying specific speech and text representation models (Whisper, wav2vec 2.0, XLS-R, HuBERT, XLM-R, LaBSE), the problem of cascading errors, the distinction between direct and indirect intent, the role of conversational context, and candidate datasets (MInDS-14, MASSIVE, Skit-S2I, Banking77). The next phase will focus on multimodal fusion strategies and identifying or extending datasets suitable for multilingual, context-aware, indirect intent detection.